Examiner only

10. The table below gives data about some objects in our solar system.

Object	Mass (units)	Diameter (km)	Length of day (hours)	Year length (days)	Orbital speed (km/s)	Mean temperature (°C)	Distance from Sun (units)
Mercury	0.330	4 879	4 222.6	88	47.4	167	0.39
Venus	4.87	12 104	2 802	225	35	464	0.72
Earth	5.97	12 756	24	365	29.8	15	1.00
Moon	0.073	3 475	708.7		1	–20	
Mars	0.642	6 792	24.7	687	24.1	–65	1.52
Jupiter	1 898	142 984	9.9	4 331	13.1	–110	5.20
Saturn	568	120 536	10.7	10 747	9.7	–140	9.54
Uranus	86.8	51 118	17.2	30 589	6.8	–195	19.18
Neptune	102	49 528	16.1	59 800	5.4	–200	30.06
Pluto	0.0146	2 370	163.3	90 560	4.7	–225	39.53

(a) (i) Use the information from the table to **tick (✓)** the **two** correct statements below. [2]

Neptune is hotter than the Moon. ☐

The mean temperature of the Moon is 5 degrees less than the Earth. ☐

A year on Earth is about 4 times longer than a year on Mercury. ☐

Mercury orbits the Sun with a speed around 10 times greater than Pluto. ☐

(ii) Pluto was once considered to be a planet but is now classed as a dwarf planet. Use the data in the table to suggest a reason for the change. [1]

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(iii) Ceres is another dwarf planet and it is the only dwarf planet located in the asteroid belt. Estimate its distance from the Sun. [1]

Distance from the Sun = ... units

(b) Elin concludes that rocky planets with the greatest mass have the shortest day length. Explain the extent to which the data supports this conclusion. [2]

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END OF PAPER